Proceedings Track

Truth Has a Boundary. A Machine-Verified Topological Constraint on Models That Separate Truth from Falsehood

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Abstract

Probing studies show that truth is easy to read off a language model's activations. Linear probes find truth directions. True and false statements separate in representation space. We ask the reverse question. What does the shape of that space require of any model that answers over it? The model is any continuous map from a connected representation space to answers and confidences. Truth is a continuous signed field with the truth boundary as its zero set. Our central result is a coupling theorem. If the model ever answers truly and ever answers falsely, and its confidence separates true answers from false ones, then some query must carry confidence exactly at the decision threshold while its answer sits exactly on the truth boundary. The theorem mentions no safety guarantee, so it does not depend on how a guarantee treats its own threshold. Consequences follow per policy. A guarantee that includes its threshold is impossible. A guarantee that excludes its threshold survives but is silent at the coupled query. Under Lipschitz regularity the coupled point sits inside an ambiguity tube of computable width, and any feasible system must carry strictly positive truth-side slack. For nonzero linear probes the truth boundary is exactly an affine hyperplane of codimension one. We further prove a symmetry version that needs no coverage assumption, a discrete analysis that prices the removal of topology, and multi-turn and stochastic extensions. The stochastic dichotomy is where falsehood is derived. Every theorem in this paper is machine-verified in Lean 4 against Mathlib, with no sorry placeholders, and the main results reduce to the three standard kernel axioms. An appendix observes the predicted coupling in five small language models, with the model's own confidence signal and against null baselines, and an adversarial attempt to train away the slack plateaus strictly above zero.

Keywords: representational geometry, topology, uncertainty, calibration, intermediate value theorem, Borsuk-Ulam, formal verification, Lean

1. Introduction

Truth is geometrically simple inside language models. Linear probes separate true from false statements in activation space (Marks and Tegmark, 2024; Azaria and Mitchell, 2023). Latent knowledge can be found without supervision (Burns et al., 2023). Steering activations along learned directions makes models more truthful (Li et al., 2023; Zou et al., 2023). All of this work treats truth as a continuous, decodable field over representation space.

We take that picture seriously as geometry and ask what follows. Suppose truth is a continuous scalar field on a connected representation space, and the model's answers and confidences are continuous fields on the same space. These assumptions mention no architecture, training method, or scale, and they apply to any model whose fields are defined and continuous over the stated domain. What they yield is a coupling theorem. If the model ever answers truly and ever answers falsely, and its confidence separates true answers from false ones, then some query carries confidence exactly at the threshold while its answer sits exactly on the truth boundary.

The proof is short. The value of the theorem lies in what it does not mention. It contains no safety guarantee and no convention about thresholds, so every policy question becomes a corollary. An inclusive guarantee is impossible. An exclusive guarantee survives but says nothing at the one point that topology guarantees to exist. Slack buys safety, and we quantify the trade. Under Lipschitz regularity the coupled point sits inside an ambiguity tube of computable width, and for the nonzero linear probes used in practice the truth boundary is an affine hyperplane.

Contributions.

1. **A boundary coupling theorem** (Section 4). Coverage plus continuity yield an exact truth-boundary point, truth separation gives it threshold confidence, and the combined theorem is convention-free, with a taxonomy of corollaries classifying each policy.

2. **The shape of the boundary** (Section 5). The truth boundary is closed. Every continuous path from a true answer to a false answer crosses it. For a nonzero linear probe it is an affine translate of the probe's kernel, of dimension exactly $d-1$.

3. **The ambiguity tube** (Section 6). With Lipschitz confidence, decisive regions are separated by a corridor of width at least $2\gamma/L$, and any system with band separation and a threshold-inclusive guarantee must have strictly positive truth-side slack.

4. **A symmetry version** (Section 7). If semantic negation acts freely and continuously and the truth field is odd under it, a boundary point exists with no coverage assumption at all.

5. **A discrete analysis** (Section 8) that prices the removal of topology, and **extensions** (Section 9) to multi-turn and stochastic models.

6. **Full machine verification** (Section 10). Every theorem in this paper corresponds to a named, sorry-free Lean 4 result checked against Mathlib.

What the contribution is. The mathematics is elementary, and the contribution is not technical difficulty. It is a framing in which exact threshold uncertainty is a topological property of representation space, stated in the vocabulary of the probing literature. It is precision, since the theorems name every assumption in play, so debate moves from proofs to premises. And it is verification, since every statement is machine-checked and the premises are the only thing left to argue about. The theorems do not say models err often, and they assign no probability to failure. The coupled point of the deterministic results lies on the truth boundary, neither strictly true nor strictly false, and derived falsehood appears only in the stochastic dichotomy.

2. Related work

Hallucination inevitability. Kalai and Vempala (2024) show statistically that calibrated models must hallucinate on facts seen once in training, Kalai et al. (2025) extend this line, and Xu et al. (2024) give a computability-theoretic inevitability argument. Our mechanism is topological, lives in representation space, localizes the conclusion at the truth boundary, and is more modest, since the deterministic theorems yield uncertainty rather than confident falsehood.

Geometry of truth in representations. Linearly decodable truth (Marks and Tegmark, 2024), unsupervised latent-knowledge discovery (Burns et al., 2023), lie detection from internal states (Azaria and Mitchell, 2023), and causal steering along truth directions (Li et al., 2023; Zou et al., 2023) all model truth as a continuous decodable field on activation space, and the linear representation hypothesis formalizes concepts as directions in that space (Park et al., 2024). That is exactly our hypothesis on δ, and confidence signals (Farquhar et al., 2024; Kadavath et al., 2022) are treated similarly. Our results give this program a frontier, since a probe that exactly separates truth over a connected domain with both truth values realized must have an exact-uncertainty point on its own boundary.

Topology, geometry, and calibration. The manifold hypothesis (Fefferman et al., 2016) and the measured geometry of embedding spaces (Cai et al., 2021) motivate connected-domain modeling. Topological analyses of networks (Naitzat et al., 2020; Hensel et al., 2021) and geometric deep learning (Bronstein et al., 2021) engineer or analyze such structure. We derive constraints from it. The symmetry section is a one-dimensional relative of the Borsuk-Ulam theorem (Borsuk, 1933; Matoušek, 2003). Our separation condition is not statistical calibration in the sense of Guo et al. (2017), and Remark 6 spells out the difference.

Formalized machine learning theory. To our knowledge this is the first fully machine-verified impossibility-style result on model truthfulness at the level of representation geometry (de Moura and Ullrich, 2021; mathlib Community, 2020).

3. The setting

$\mathcal{X}$ is a topological space of queries as the model represents them, such as prompt embeddings or residual-stream states. $\mathcal{A}$ is a topological space of answers. Our standing assumption is simple.

(T) $\mathcal{X}$ is connected and nonempty.

Assumption (T) deserves care. Ambient embedding spaces are standardly modeled as $\mathbb{R}^d$ or connected submanifolds (Fefferman et al., 2016; Cai et al., 2021), which satisfy (T). But the set of representations a token-driven model reaches may be disconnected. When guarantees range over the ambient space, as they implicitly do whenever a probe or steering vector is applied at arbitrary points, (T) holds. Section 8 analyzes what remains without it.

Definition 1 (Model field) *A* model *is a map $F = (a, c)$ from $\mathcal{X}$ to $\mathcal{A} \times \mathbb{R}$. Here $a(x)$ is the answer at representation x and $c(x)$ is its scalar confidence. F is* continuous *if both component maps are.*

Definition 2 (Truth field) *A* truth field *is a continuous map δ from $\mathcal{X} \times \mathcal{A}$ to $\mathbb{R}$, read as a signed truth-distance. Negative means the answer is strictly true. Positive means strictly false. Zero is the* truth boundary*. The* realized truth field *of a model is $\delta_F(x) = \delta(x, a(x))$.*

A linear probe $w^\top x - b$ on activations (Marks and Tegmark, 2024; Burns et al., 2023) is a candidate continuous δ_F, and its decision boundary is the truth boundary. Probing experiments fit classifiers on sampled representations, and the zero set of a probe is the probe's classification boundary. Whether it coincides with semantic truth away from those samples is an empirical premise, not a theorem. Our results hold under that premise.

Definition 3 (Truth-separating at threshold τ) *Fix a threshold τ, a truth slack $\varepsilon \geq 0$, and a confidence margin $\gamma \geq 0$. F is* (ε, γ)-separating at τ *if for all x two implications hold. If $\delta_F(x) < -\varepsilon$ then $c(x) > \tau + \gamma$. If $\delta_F(x) > \varepsilon$ then $c(x) < \tau - \gamma$. The truth slack is measured in units of δ_F and the margin in units of c, so the condition survives independent rescaling of either field. We say* exactly separating *when both are zero.*

Definition 4 (Covering) *F is* ε-covering *if some query has $\delta_F < -\varepsilon$ and some query has $\delta_F > \varepsilon$. At $\varepsilon = 0$ this says the model is somewhere strictly right and somewhere strictly wrong.*

Definition 5 (Guarantees) *A* threshold-inclusive guarantee *says that $c(x) \geq \tau$ implies $\delta_F(x) < 0$. A* threshold-exclusive guarantee *says that $c(x) > \tau$ implies $\delta_F(x) < 0$.*

Covering is the weakest condition, since any model that is ever right and ever wrong satisfies it. Exact separation idealizes a perfect truth probe used as confidence, the guarantees idealize hallucination mitigation, the coupling results hold at any threshold, and the biconditional variants sit at one half.

Remark 6 (This is not statistical calibration) *Statistical calibration asks that confidence match accuracy on average (Guo et al., 2017). Separation is pointwise and much stronger, since confidence classifies truth. It is also one-way by construction, and the coupling theorem and its corollaries use only the two truth-to-confidence implications. The* biconditional *variant adds the converses, so the sign of δ_F and the side of the threshold determine each other. The symmetry and discrete results use it, and it implies the one-way pair (Lean `strictCalibrated_to_epsCalibrated_zero_half`).*

Remark 7 (On continuity) *Continuity idealizes trained networks, since softmax confidences and embedding maps are continuous by construction while hard argmax decoding is not. Dropping continuity is an escape, and Section 8 prices it.*

4. The boundary coupling theorem

Topology enters through one theorem, which needs no confidence at all.

Theorem 8 (Boundary existence) *Let $\mathcal{X}$ satisfy (T). Let the answer map and the truth field be continuous, and let F be 0-covering. Then some x_0 has $\delta_F(x_0) = 0$.*

Proof The realized truth field δ_F is continuous as a composite, negative somewhere and positive somewhere, so the intermediate value theorem on the connected space $\mathcal{X}$ gives an exact zero. This is Lean theorem `model_truth_boundary_nonempty`, and the IVT step is Mathlib's `IsPreconnected.intermediate_value2`. ■

Theorem 9 (Boundary coupling) *Let $\mathcal{X}$ satisfy (T). Let c be continuous, and let F be 0-covering and exactly separating at τ. Then some x_0 satisfies both*

$$c(x_0) = \tau \qquad \text{and} \qquad \delta_F(x_0) = 0.$$

Proof Separation converts the coverage witnesses into confidence witnesses above and below τ, and the IVT gives x_0 with $c(x_0) = \tau$. If $\delta_F(x_0)$ were negative, separation would give $c(x_0) > \tau$, and if positive, $c(x_0) < \tau$. Both fail, so $\delta_F(x_0) = 0$. This is Lean theorem `boundary_coupling`. ■

The theorem mentions no guarantee, so no threshold convention affects it. We call such an x_0 a *coupled point*. The model carries exact threshold confidence where truth changes sign, and what that means for safety is a matter of policy, with each policy a corollary.

Corollary 10 (Inclusive guarantees are impossible) *Add a threshold-inclusive guarantee to the hypotheses of Theorem 9. Then the hypotheses are jointly contradictory. This is the trilemma, since coverage, exact separation, and an inclusive guarantee cannot jointly hold.*

Corollary 10 is Lean theorem `closed_guarantee_impossible`, and its fixed-threshold biconditional version is `hallucination_trilemma`.

Corollary 11 (Exclusive guarantees go silent) *Add a threshold-exclusive guarantee instead. No contradiction follows, but the coupled point satisfies $c(x_0) = \tau$, so the guarantee's antecedent fails exactly there. The system owns a query about which its guarantee says nothing, and that query sits on the truth boundary. This is Lean theorem `open_guarantee_silent`.*

Remark 12 (The convention question) *A natural objection says that defining the guarantee with a strict inequality dissolves the contradiction, so the impossibility is an artifact of convention. The taxonomy above is our answer. The coupling theorem is convention-free, and the convention only selects which corollary describes the coupled point, a contradiction under an inclusive guarantee and silence under an exclusive one. The coupled point exists under either convention.*

5. The shape of the boundary

The truth boundary is not an isolated curiosity. It has structure, and for the probes used in practice it is a hyperplane.

Theorem 13 (The boundary is closed) *For any topological $\mathcal{X}$ and $\mathcal{A}$, continuous δ and answer map, the set $\{x \mid \delta_F(x) = 0\}$ is closed in $\mathcal{X}$. This is Lean theorem `model_truth_boundary_isClosed`.*

Theorem 14 (Every path crosses) *Let p be a continuous path in $\mathcal{X}$ from a query where the model answers strictly truly to a query where it answers strictly falsely. Then some point on the path lies exactly on the truth boundary. This is Lean theorem `path_crosses_truth_boundary`.*

The boundary therefore separates the true region from the false region. For linear probes we can say exactly what it is.

Theorem 15 (Linear probes have hyperplane boundaries) *Let f be a nonzero linear probe on a d-dimensional real space, with offset b. Then the boundary set $\{x \mid f(x) = b\}$ is an affine translate of the kernel of f, and that kernel has dimension exactly $d-1$. These are Lean theorems `linear_probe_boundary_coset` and `linear_probe_boundary_dim`, the latter by rank-nullity.*

The truth boundary of practical probes is therefore a codimension-one affine hyperplane, and the coupling theorem places a threshold-confidence answer on it. Truth has a boundary surface, not only a boundary point.

6. The ambiguity tube

Exact separation is an idealization, so we quantify what slack changes. Two Lipschitz facts come first and need only a metric space.

Theorem 16 (Corridor width) *Let c be L-Lipschitz with $L > 0$ and let $\gamma > 0$. Then any query with $c \geq \tau + \gamma$ and any query with $c \leq \tau - \gamma$ are at distance at least $2\gamma/L$. This is Lean theorem `confidence_gap_width`.*

Theorem 17 (Ambiguity tube) *Let c be L_c-Lipschitz and δ_F be L_δ-Lipschitz, and let x_0 be a coupled point from Theorem 9. Then every query within distance r of x_0 has truth margin at most $L_\delta r$ and confidence within $L_c r$ of the threshold. This is Lean theorem `ambiguity_tube`, with the margin bound alone as `truth_margin_tube`.*

The coupled point is therefore not one pathological query. Around it sits a tube of near-ambiguous queries controlled by the two Lipschitz constants, and decisive regions cannot approach closer than the corridor width. Slack changes what the coupled point looks like, and we can say exactly how.

Theorem 18 (Approximate coupling) *Let $\mathcal{X}$ satisfy (T), let c be continuous, and let F be ε-covering, (ε, γ)-separating at τ, and carry a threshold-inclusive guarantee. Then some x_0 has $c(x_0) = \tau$ and $-\varepsilon \leq \delta_F(x_0) < 0$. This is Lean theorem `two_slack_approx_coupling`, with the exact case as `exact_from_approx`.*

Theorem 19 (Truth slack must be positive) *Under the hypotheses of Theorem 18, the truth slack satisfies $\varepsilon > 0$ for every margin $\gamma \geq 0$. Zero truth slack is infeasible. This is Lean theorem `truth_slack_must_be_positive`.*

Slack is therefore a geometric budget, and positivity attaches to the truth side alone, whatever the confidence margin. We propose reading the infimum of feasible truth slack as the system's truth-boundary width, and the relative boundary width measured in Appendix C gives a dimensionless proxy for this quantity. The corridor is also clean. Under the guarantee, no query with confidence between τ and $\tau + \gamma$ sits exactly on the truth boundary (Lean theorem `no_boundary_in_upper_band_two_slack`).

7. Symmetry replaces coverage

The coupling theorem uses coverage to seed the IVT, and symmetry can replace it entirely. Suppose semantic negation acts on $\mathcal{X}$ as a continuous involution σ with no fixed point, and the realized truth field is odd under it, idealizing the observation that negation approximately reflects truth directions (Marks and Tegmark, 2024; Burns et al., 2023).

Theorem 20 (Odd fields vanish) *Let $\mathcal{X}$ satisfy (T) with such an action σ, and let g be continuous and odd under σ. Then g has a zero. The proof picks any x and applies the IVT between x and σx. This is Lean theorem* `antipodal_odd_has_zero`*.*

This is the scalar shadow of the Borsuk-Ulam theorem (Borsuk, 1933; Matoušek, 2003). It needs no sphere, since any connected space with a free continuous $\mathbb{Z}/2$ action supports it. Two refinements make the section robust and concrete.

Theorem 21 (Robust and concrete oddness) *Let $\mathcal{X}$ satisfy (T), let σ map $\mathcal{X}$ to itself, and let g be continuous with $|g(\sigma x) + g(x)| \leq \varepsilon$ for all x. Then some x has $|g(x)| \leq \varepsilon/2$. Moreover, on the unit sphere of $\mathbb{R}^d$ with $d \geq 2$ and negation acting as the antipodal map, every continuous field odd under the antipode vanishes somewhere on the sphere. These are Lean theorems* `approx_odd_near_zero` *and* `sphere_odd_has_zero`*.*

The near-boundary point degrades linearly with the oddness defect, and the sphere case is, up to a learned affine map, where layer normalization places its normalized states, with negation as a candidate antipode.

Theorem 22 (Antipodal coupling) *Let $\mathcal{X}$ satisfy (T) with a free continuous involution σ, let F be continuous with biconditional separation at $\frac{1}{2}$, and let δ_F be odd. Then some x has $c(x) = \frac{1}{2}$ and $\delta_F(x) = 0$. This is Lean theorem* `antipodal_hallucination_trilemma`*.*

No coverage hypothesis appears, and a model answering a negation-closed family of questions meets the truth boundary by symmetry alone. Equivariance is usually a design virtue (Bronstein et al., 2021), and here it is the engine of the obstruction.

8. What leaving the continuum costs

Token spaces are finite, and one may object that connectedness does illegitimate work. The discrete analysis makes the objection and its price exact. Without topology the boundary witness must be assumed rather than derived.

Theorem 23 (Discrete impossibility) *Let $\mathcal{X}$ and $\mathcal{A}$ be arbitrary sets with arbitrary F and δ. Suppose F has biconditional separation at $\frac{1}{2}$ and some query has $\delta_F = 0$. Then a threshold-inclusive guarantee fails. Moreover, under separation, such a guarantee is possible exactly when no boundary query exists. These are Lean theorems* `boundary_question_impossible` *and* `faithful_iff_boundary_free`*.*

The boundary witness is assumed here, and it is exactly what connectedness derived for free in Theorem 8. What survives without any witness assumption is a crossing.

Proposition 24 (Discrete sign change) *Let g map $\{0,\dots,n\}$ to $\mathbb{R}$ with $g(0) < 0 < g(n)$. Then some i has $g(i) < 0 \leq g(i+1)$. This is Lean theorem `discrete_sign_change`.*

Along any token-space path from true to false, the realized truth field crosses zero between adjacent steps. This yields a straddling pair rather than a boundary state, hence no contradiction on its own. The continuum is precisely what upgrades a crossing into a witness.

9. Products and randomness

Multi-turn models. A T-turn model sees a history in $\mathcal{X}^T$, products of connected spaces are connected, and the connectedness arguments apply verbatim to history-dependent models. The Lean artifact transports the trilemma this way (`multi_turn_history_dependent`). Conversation context moves the truth boundary into history space rather than removing it.

Stochastic models, where falsehood appears. Applied to the expected fields, the coupling theorem yields a query with expected confidence one half and expected truth-distance zero (Lean theorem `stochastic_coupling_expected`). A dichotomy converts expectation into sampling behavior.

Theorem 25 (Stochastic dichotomy) *Let (Ω, μ) be a probability space and let c, d be real functions on Ω with d integrable and of mean zero. Suppose sample faithfulness holds almost everywhere, meaning $c \geq \frac{1}{2}$ implies $d < 0$, and suppose $\mu\{c \geq \frac{1}{2}\} > 0$. Then $\mu\{d > 0\} > 0$. This is Lean theorem `stochastic_dichotomy`.*

A model faithful on almost every sample and confident with positive probability at a coupled point must emit strictly false answers with positive probability. Randomization spreads the boundary across samples, and this is where falsehood is derived.

10. Machine verification

Every theorem and proposition in this paper corresponds to a named, sorry-free Lean 4 result (de Moura and Ullrich, 2021) checked against Mathlib (mathlib Community, 2020), both pinned at v4.28.0. The anonymized `HallucinationProofs` artifact ships seventeen source files, `lake build` completes with zero errors and no `sorry`, and axiom audits reduce each main theorem to exactly the three standard kernel axioms. Table 1 in Appendix A maps every result to its identifier, with `hallucination_master_theorem` bundling the core impossibility. The formal statements are slightly more general than the prose, and the prose claims nothing beyond what is formalized.

11. Discussion

The coupling theorem rests on connectedness, continuity, coverage, and separation, and the trilemma adds a threshold-inclusive guarantee. Each hypothesis can be relaxed, and each relaxation is a design stance. Corollary 11 addresses the stance that excludes the threshold, whose guarantee survives but owns a silent boundary point.

Disconnect the space. Abstention breaks the argument at the IVT step, since declining to answer on an open region deletes it from the domain and can disconnect the true region from the false region. Refusal training, on this reading, is connectedness surgery. Under the symmetry version the cut must separate every representation from its negation image, so piecemeal refusal is not enough.

Quantify only over reachable queries. The coupled point may be an ambient activation that no real prompt produces, and a guarantee quantified only over reachable representations is then untouched. The objection is fair, and it is the domain version of disconnecting the space. But probing and steering pipelines apply their readouts at interpolated and edited activations, which is deployment over the ambient space, and a guarantee over the uncertified reachable set inherits every unknown of that set. Restricting the domain is a coherent stance if stated as one.

Give up coverage or continuity. A model that never answers strictly falsely escapes. That is vacuous for general-purpose systems and meaningful for retrieval-grounded ones restricted to verified content. Dropping continuity also escapes, at the price computed in Section 8, where only the adjacent-crossing statement survives unconditionally.

Separate to a band. Theorems 18 and 19 make slack the correct optimization target. A feasible system must carry strictly positive truth slack, with guarantees available only outside the band between τ and $\tau + \gamma$. The infimum of feasible truth slack is measurable, and we propose it as a diagnostic.

Implications for probing and steering. Truth-probe pipelines posit a continuous δ_F, and steering moves representations relative to its zero set (Li et al., 2023; Zou et al., 2023). Completing such a pipeline over a connected domain with both truth values realized yields an exact-uncertainty point on the probe's own boundary, a hyperplane for linear probes, and mitigation effort should go there.

Limitations and outlook. The results are existence statements, so nothing follows about error rates. Deterministic conclusions concern boundary points, with Theorem 25 as the stochastic exception. The truth field idealizes probe behavior, and Theorem 21 prices oddness. Appendix C reports empirical checks on five small models. Next steps are higher Borsuk-Ulam machinery for jointly odd axes and nonlinear probe boundaries (Engels et al., 2025), in brains and in machines alike. The geometry of representation space can prove theorems, and machines can check them.

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Appendix A. Formal statement conventions

The Lean formalization states results over arbitrary topological spaces `Q` and `A`, with a `ConnectedSpace` instance on `Q`, a model `M` from `Q` to `A x R`, and a truth field `delta` on `Q x A` (rendered here in ASCII). The realized truth field is

```
fun q => delta (q, (M q).1)
```

and confidence is `(M q).2`. The main conditions are named as follows, and some identifiers retain historical names.

- `TwoSlackSeparating M delta c eps gamma` is separation at threshold `c` with truth slack `eps` and confidence margin `gamma`. The equal-slack case is `EpsCalibrated M delta c eps`, definitionally (`epsCalibrated_iff_twoSlack`).
- Covering with slack `eps` is `EpsCovering M delta eps`.
- The biconditional variant at threshold one half is `StrictCalibrated M delta`.
- The threshold-inclusive guarantee at threshold one half is `TrilemmaFaithful M delta`. The general threshold version appears inline in the theorems that use it.

The embedding-space instantiation uses `EuclideanSpace R (Fin d)`. The quantitative results use `PseudoMetricSpace` and explicit Lipschitz hypotheses, and the hyperplane results use `LinearMap` with rank-nullity.

Table 1: Every result in the paper and its machine-checked counterpart.

Result	Lean identifier
Thm. 8	model_truth_boundary_nonempty
Thm. 9	boundary_coupling
Cor. 10	closed_guarantee_impossible, hallucination_trilemma
Cor. 11	open_guarantee_silent
Thm. 13	model_truth_boundary_isClosed
Thm. 14	path_crosses_truth_boundary
Thm. 15	linear_probe_boundary_coset, linear_probe_boundary_dim
Thm. 16	confidence_gap_width
Thm. 17	ambiguity_tube, truth_margin_tube
Thm. 18	two_slack_approx_coupling, exact_from_approx
Thm. 19	truth_slack_must_be_positive
Upper-band exclusion	no_boundary_in_upper_band_two_slack
Thm. 20	antipodal_odd_has_zero
Thm. 21	approx_odd_near_zero, sphere_odd_has_zero
Thm. 22	antipodal_hallucination_trilemma
Thm. 23	boundary_question_impossible, faithful_iff_boundary_free
Prop. 24	discrete_sign_change
Multi-turn	multi_turn_history_dependent
Stochastic expected-value coupling	stochastic_coupling_expected
Thm. 25	stochastic_dichotomy
Bundled master theorem	hallucination_master_theorem

Appendix B. Core proof term

The Lean proof of the coupling theorem, lightly annotated and rendered in ASCII.

```
theorem boundary_coupling
    {Q A : Type*} [TopologicalSpace Q] [ConnectedSpace Q]
    [TopologicalSpace A]
    (M : Q -> A x R) (delta : Q x A -> R) (c : R)
    (hconf : Continuous (fun q => (M q).2))
    (hcov : EpsCovering M delta 0)
    (hcal : EpsCalibrated M delta c 0) :
    exists q0, (M q0).2 = c /\ delta (q0, (M q0).1) = 0 := by
  obtain <<qt, ht>, <qf, hf>> := hcov
  -- separation turns coverage witnesses into confidence witnesses
  have hct : (M qt).2 > c + 0 := hcal.1 qt ht
  have hcf : (M qf).2 < c - 0 := hcal.2 qf hf
  have hct' : (M qt).2 > c := by linarith
  have hcf' : (M qf).2 < c := by linarith
  -- IVT on the confidence field
  obtain <q0, _, hq0> :=
    isPreconnected_univ.intermediate_value2
      (mem_univ qf) (mem_univ qt)
      hconf.continuousOn continuous_const.continuousOn
      (le_of_lt hcf') (le_of_lt hct')
```

```
refine <q0, hq0, ?_>
-- at conf = c, separation excludes both signs
by_contra hne
rcases lt_or_gt_of_ne hne with hlt | hgt
. have hd : delta (q0, (M q0).1) < -0 := by simpa using hlt
  have := hcal.1 q0 hd
  linarith
. have := hcal.2 q0 hgt
  linarith
```

Appendix C. Empirical illustration on small language models

The theorems are about idealized fields, so we checked whether their predictions are visible in real models. We used five open language models, namely GPT-2 with 124M parameters, Pythia-410M, Qwen2.5-0.5B, Phi-1.5 with 1.3B parameters, and Qwen2.5-1.5B, the first three below one billion parameters. The data are the matched city statements and their negations from the geometry-of-truth datasets of Marks and Tegmark (2024), giving 1496 true and false statement pairs such as "The city of Paris is in France" and its negation.

Setup. We embed each raw statement at its final token and select the layer whose probe has the best validation accuracy. The truth field δ_F is a linear logistic probe trained on one split of cities. The confidence field c is a second linear logistic head trained on a disjoint split with a different seed, so the two fields are independently trained readouts of the same representation space. This is the canonical construction the theory models, since a probing pipeline that uses a truth readout as a confidence signal is exactly a truth-separating c. We then interpolate hidden states between each matched pair, from the true statement to its negation, and record both fields along each path. We keep the paths whose endpoints the probe classifies correctly, which is 83 paths for GPT-2, 104 for Pythia-410M, 137 for Qwen2.5-0.5B, 66 for Phi-1.5, and 148 for Qwen2.5-1.5B. Test cities are held out from both training splits.

Findings. Table 2 and Figure 1 summarize. First, the premises hold approximately. Linear probes separate truth with accuracy 0.72 to 1.00, and only the two Qwen models are well above chance when asked the question verbally, at 0.71 and 0.96 against 0.48 to 0.56 for the rest, which replicates the known gap between represented and verbalized truth. Second, every kept path shows the adjacent sign change predicted by the discrete result. Third, confidence is pinned near the threshold at the boundary. The mean value of c at the probe's zero crossing is 0.52, 0.51, 0.49, 0.43, and 0.54 for the five models, against decisive values near 0 and 1 at the endpoints. Fourth, the two independently trained fields couple. The median gap between the zero of δ_F and the threshold crossing of c tracks probe accuracy monotonically across all five models, from 0.31 of the path for Phi-1.5, whose probe is weakest, to 0.05 for Qwen2.5-1.5B, whose probe is strongest. The ordering follows separation quality and not parameter count alone, since Phi-1.5 outweighs every model except Qwen2.5-1.5B yet is the weakest separator on this domain. A sweep of all 24 of its layers finds no linear probe above 0.73, so the weakness is a property of the model on this domain and not of the layer choice. The two Qwen models give a within-family scale comparison, and there scale helps through separation quality, raising the probe from 0.96 to 1.00 and halving the gap from 0.10 to 0.05. Fifth, the relative boundary width, the median $|\delta_F|$ over queries whose confidence lies in $[0.4, 0.6]$ divided by the endpoint scale, runs from 0.37 for the weakest probe to 0.11 for the strongest. Sixth, the kNN graph of held-out activations places true and false statements in one shared main component for all five models, so truth values are not separated into different components, though three models show a second component. A diagnostic below identifies what that component is.

The model's own confidence. The design above trains both fields on truth, so nearby boundaries could reflect shared fitting. We therefore repeat the experiment with a confidence field the model supplies itself, with no trained head, read out through unembedding weights fixed by pretraining before our data existed. We capture the final block's last-token state under a question template, interpolate between matched pairs as before, patch the interpolated state back into the final block, and read the model's own probability of answering true over the two answer tokens. The truth probe is trained as before on a disjoint split of the same states. The question-template states change which endpoint pairs the probe classifies correctly, giving 53, 84, 137, 80, and 147 kept paths here. The endogenous field satisfies the separation premise on a path when it is decisive at both endpoints, meaning its confidence lies on the correct side of one half at both. Table 3 and Figure 2 report what follows. In Qwen2.5-1.5B the premise holds on 146 of 147 paths with median gap 0.08, and in Qwen2.5-0.5B on 128 of 137 paths with median gap 0.13, both against null gaps of 0.40 and above. In Pythia-410M the premise fails on every path, since the model answers true regardless of the statement, and the theory predicts nothing. GPT-2 sits between, with 23 of 53 paths decisive and a gap that does not separate from the random-head null. Phi-1.5 is endpoint decisive on 53 of 80 paths, but its verbal accuracy is 0.56 and its gap stays at null level, so endpoint decisiveness without verbal capability does not amount to separation. Coupling with the model's own uncertainty appears cleanly where the separation premise holds throughout, and never where it fails.

Null baselines. To calibrate the gaps, we replace the trained confidence head with heads trained on permuted labels and with random directions, twenty seeds each, evaluated on the same kept paths as the trained heads (Table 3). The trained-head gaps of 0.25, 0.20, 0.10, 0.31, and 0.05 fall below the null medians for all five models, and below the entire null range for Pythia-410M and both Qwen models. For GPT-2 and Phi-1.5 the observed gaps lie inside the random-head null range, so the trained-head evidence is weak there on its own. Bootstrap intervals over paths, with ten thousand replicates, make the gaps precise. The 95 percent intervals for the median trained-head gap are 0.175 to 0.325, 0.15 to 0.287, 0.10 to 0.125, 0.175 to 0.375, and 0.05 to 0.075 across the five models, with 0.10 to 0.15 for Qwen2.5-0.5B's endogenous field and 0.075 to 0.10 for Qwen2.5-1.5B's. Every interval upper bound sits below the corresponding null median except the endogenous cases of GPT-2 and Phi-1.5.

A second dataset. The same pipeline runs on the Spanish-English translation statements from the same source, 354 matched pairs, with splits by statement since no city structure exists. Both Qwen models replicate the cities result, with probe accuracies 0.98 and 0.94 and median crossing gaps 0.05 and 0.10. Pythia-410M's probe drops to 0.70 on this harder set and its gap rises to null level, and Phi-1.5's probe reaches 0.71 with its gap at null level too. GPT-2's probe is at chance there, with too few valid paths to score. Coupling again appears exactly where separation holds.

What the second component is. The kNN check deserves one more look, since three models showed a second component holding half the points. The diagnostic answer is clean. The components split exactly by surface form. One holds the positive statements, the other the negations, and each contains true and false statements in equal measure. What separates activations is syntax, not truth, and both truth values coexist inside each component. The interpolation paths are line segments in the ambient space, which are connected regardless of how the data clusters.

Measuring the oddness defect. The symmetry section rests on negation acting oddly on the truth field, and the matched pairs measure how odd it is. The defect $|\delta_F(x)+\delta_F(\sigma x)|$ relative to the endpoint scale has median 0.40, 0.43, 0.30, 0.53, and 0.17 for GPT-2, Pythia-410M, Qwen2.5-0.5B, Phi-1.5, and Qwen2.5-1.5B, smallest for the strongest model. Exact oddness fails in real representations, which is why Theorem 21 carries the approximate version. Its near-boundary point survives any defect, and empirically every kept path crosses zero regardless. The width column of Table 2 also gives the slack diagnostic of Section 6 an empirical proxy, between 0.11 and 0.37 of the endpoint scale on these models.

Nonlinear probes. The linear truth field crosses zero along a linear path by construction, so we replace both fields with one-hidden-layer MLPs trained on the same splits. Nothing about the path behavior is then guaranteed, the boundary is no longer a hyperplane, and multiple crossings become possible. The nonlinear probes reach 0.79, 0.87, 0.97, 0.80, and 1.00, and give 124, 121, 144, 104, and 149 kept paths. Every kept path still crosses zero, and at least 98 percent of paths in every model cross exactly once, so the nonlinear boundary is still met transversally along these paths. The coupling tightens or holds in every model, with median gaps 0.15, 0.20, 0.075, 0.175, and 0.05, and confidence at the crossing averaging 0.42, 0.50, 0.42, 0.38, and 0.49. The coupling is therefore not an artifact of linearity.

An adversarial falsification attempt. The strongest test of an impossibility result is a maximally empowered attempt to violate it. We freeze the truth field and train a confidence head directly on the forbidden objective, confidence above one half wherever the field says true and below one half wherever it says false or boundary, with the guarantee side weighted five times harder. The adversary trains on all non-test statements plus fresh continuum interpolants every epoch, and is evaluated on 21,000 to 30,000 held-out path points per model. Theorem 19 predicts three things. The achieved truth slack must stay positive, the residual failures must localize at the boundary, and slack must trade against guarantee violations. All three appear. The achieved slack falls and then plateaus between 0.22 and 0.91 of the endpoint scale across the five models, exactly flat over the final fifteen epochs for Qwen2.5-0.5B, and never reaches zero. The median depth of failing points shrinks from 0.40 or more to between 0.04 and 0.11, so failures concentrate in the tube around the boundary. And the guarantee stays violated on a small fraction of points throughout. The theorem forbids only exactly zero slack for any fixed continuous head, so a stronger adversary could push the floor lower, and Theorem 16 prices that escape, since a sharper transition requires a proportionally larger Lipschitz constant. The observed plateau reflects the floor set by this adversary's regularity, and the localization shows what sharpening buys, a thinner tube around a boundary that does not go away.

What this does and does not show. The truth field of the main design is linear, so its sign change along a linear path is guaranteed analytically, and the nonlinear replication above removes exactly that guarantee. The empirical content lies elsewhere. The premises hold to varying degrees in real small models. The two fields are trained on disjoint data with different seeds, so their coupling at the boundary is not an artifact of shared training data, though both target truth. The endogenous experiment addresses the residual concern where the model itself can verbalize truth. The pinned confidence value lands at one half. And the boundary width is a measurable quantity that differentiates models. Code, results, and the figure are included in the supplementary material.

Table 2: Boundary coupling in five small models. Layer is the selected layer. Probe and Head are held-out accuracies of the truth probe and the confidence head. Verbal is the model's own true or false answer accuracy. Cross is the mean confidence at the probe's zero crossing. Gap is the median distance between the two fields' crossings as a fraction of the path. Width is the relative boundary width.

Model	Layer	Probe	Head	Verbal	Cross	Gap	Width
GPT-2 (124M)	9/12	0.77	0.72	0.48	0.52	0.25	0.33
Pythia-410M	18/24	0.82	0.77	0.50	0.51	0.20	0.36
Qwen2.5-0.5B	12/24	0.96	0.96	0.71	0.49	0.10	0.20
Phi-1.5 (1.3B)	18/24	0.72	0.72	0.56	0.43	0.31	0.37
Qwen2.5-1.5B	14/28	1.00	1.00	0.96	0.54	0.05	0.11

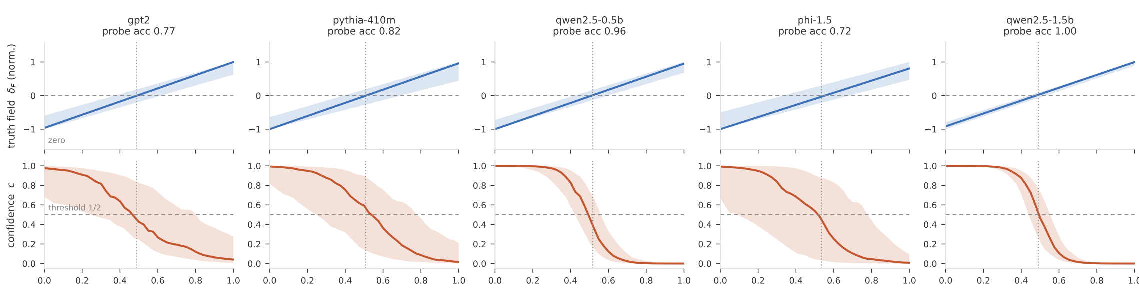

Figure 1: The predicted picture, observed. Top row shows the truth probe field along interpolation paths from a true statement to its negation, median and interquartile band over paths, normalized per path. Bottom row shows the independently trained confidence field on the same paths. The dotted vertical line marks the mean probe zero crossing. Confidence passes through its threshold of one half at the probe boundary in every model, most sharply in the strongest model.

Table 3: The model's own confidence and null calibration. Decisive counts paths where the endogenous field separates the endpoints. Endo is the median crossing gap on decisive paths. Perm and Rand are median null gaps for permuted-label and random-direction heads over twenty seeds. Heads repeats the trained-head gap from Table 2.

Model	Decisive	Endo	Perm	Rand	Heads
GPT-2 (124M)	23/53	0.23	0.48	0.33	0.25
Pythia-410M	0/84	none	0.45	0.45	0.20
Qwen2.5-0.5B	128/137	0.13	0.43	0.40	0.10
Phi-1.5 (1.3B)	53/80	0.40	0.48	0.38	0.31
Qwen2.5-1.5B	146/147	0.08	0.45	0.45	0.05

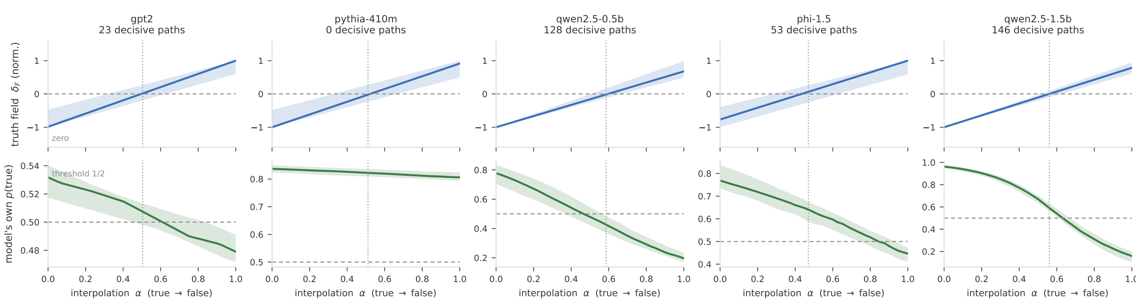

Figure 2: The model's own confidence couples to the probe boundary where the separation premise holds. Rows as in Figure 1, but the bottom row is the model's own probability of answering true, obtained by patching the interpolated final-block state, restricted to decisive paths. Qwen2.5-0.5B sweeps through one half at the probe boundary. Pythia-410M answers true everywhere, the premise fails, and no prediction follows. GPT-2 and Phi-1.5 sit between.

Appendix D. Artifact checklist

The anonymized supplementary zip contains the complete Lean project and the experiment. The Lean part includes the build configuration, the toolchain pin for Lean v4.28.0, the manifest pin for Mathlib v4.28.0, and seventeen source files. To reproduce, run `lake build`. It completes with zero errors and no `sorry`. Axiom sets of the main theorems are printed by the final-verification file and by the files numbered 13 and 14. The experiment part includes the scripts, the public datasets of Marks and Tegmark (2024), the numerical results, and the figures. The theorems themselves depend on no experiment.